📘 MEP Calculator — End-User & Organization Operations Manual

> **NOTE:** This official manual is designed for **Organization Admins, Senior Design Engineers, Junior Design Engineers, and Viewers**. It provides step-by-step instructions on tab navigation, engineering calculations, role-specific tasks, and license activation.

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🚀 1. Quick Start & License Activation Guide

🎁 14-Day Free Trial

When you first register an account, your email receives an OTP verification code. Upon logging in, you automatically receive a **14-Day Free Trial** with unrestricted access to all calculation tools.

💳 How to Purchase a Pro Plan & Paste License Key

When your 14-day trial expires (or when your organization purchases a Pro License):

```mermaid
flowchart LR
    Step1["1. Purchase License Key from Provider"] Step2["2. Org Admin Receives Key (MEP-XXX-YYY)"] Step3["3. Click 'Activate License' in Header"] Step4["4. Paste Key & Click Submit"]
```

Step5["5. Auto-Bind Org Name & Unlock Seats"]

Step1 --> Step2 --> Step3 --> Step4 --> Step5 `

Step-by-Step Activation Instructions:

1. **Locate Your License Key**: Obtain your 16-character license key (format: MEP-XXXX-XXXX-XXXX) provided to your organization admin. 2. **Open License Popup**: - Click your **Profile Name** or **Settings** icon in the top-right header. - Click the purple **"Activate License Key"** button (or wait for the automatic renewal popup). 3. **Paste Key**: Paste your key into the text field. 4. **Click Activate**: Click **"Submit / Activate"**. 5. **Automatic Organization Binding**: - Your account is immediately linked to your organization (organization_id). - Your organization name (e.g. *ABC Engineering Consultants*) automatically displays under your profile. - Your license status changes to **ACTIVE**.

🗺️ 2. Application Layout & Tab Navigation

The MEP Calculator features an intuitive sidebar navigation structure:

Sidebar Tab	Description & Key Functionality
📊 **Dashboard**	Overview of active projects, assigned teams, recent activity logs, and system announcements.
📁 **Projects**	Central hub to create new engineering projects, assign team members, and view load schedules.
🧮 **Calculators**	Complete suite of HVAC, Plumbing, Firefighting, and Electrical calculation tools.
👥 **Teams**	Create internal design teams, add engineers to teams, and link teams to specific projects.
🏢 **Organization**	*(Org Admin Only)* Manage paid user seats, view active members, and configure backup paths.
⚙️ **Settings**	Update your user profile, change password, manage shortcuts, and check for software updates.

⚡ 3. Engineering Calculators Suite (HVAC, Plumbing & Electrical)

🌀 A. HVAC (Heating, Ventilation & Air Conditioning) Tab

Calculator Name	Where to Click	What it Calculates & Computes
Duct Sizer	*Calculators → HVAC → Duct Sizer*	Calculates equal friction duct dimensions (Width x Height), air velocity (FPM), friction loss per 100 ft, and equivalent round diameter.
Preliminary Cooling Load	*Calculators → HVAC → Cooling Load*	Estimates total room tonnage (TR), sensible/latent heat gains, and CFM required per sq ft.
Fresh Air Cooling Load	*Calculators → HVAC → Fresh Air Load*	Computes outdoor air ventilation requirements based on ASHRAE 62.1 occupant density and outdoor air enthalpy.
Refrigerant Pipe Sizing	*Calculators → HVAC → Refrigerant Sizing*	Sizes suction, discharge, and liquid refrigerant lines for R410A, R32, R134a, and R407C systems.
Diffuser Selector	*Calculators → HVAC → Diffuser Selector*	Determines supply air grille and diffuser neck sizes, throw distance, and NC noise criteria levels.
Smoke Extract Fan	*Calculators → HVAC → Smoke Extract*	Calculates smoke extraction exhaust rates (CFM) for atrium and corridor smoke management.
Basement Smoke Calc	*Calculators → HVAC → Basement Smoke*	Calculates air change rates (ACH) for basement car park ventilation per NBC/BS standards.
Staircase Pressurization	*Calculators → HVAC → Stair Pressurization*	Calculates fan airflow (CFM) required to maintain positive pressure (50 Pa) across open/closed staircase doors.
Lobby Pressurization	*Calculators → HVAC → Lobby Pressurization*	Determines pressure differential fan duty for elevator lobbies and fire escape routes.
Condensate Drain Calc	*Calculators → HVAC → Condensate Drain*	Sizes FCU/AHU condensate drain pipe diameters based on cooling coil latent tonnage.
Psychrometric Chart	*Calculators → HVAC → Psychrometric*	Plots dry bulb, wet bulb, relative humidity, dew point, enthalpy, and specific volume.
Heating Load	*Calculators → HVAC → Heating Load*	Calculates winter fabric heat loss (U-value x Area x ΔT) and infiltration heat loss.
Zones & Spaces	*Calculators → HVAC → Zones*	Groups multiple thermal spaces into conditioned zones for AHU sizing.

🚰 B. Plumbing & Firefighting Tab

Calculator Name	Where to Click	What it Calculates & Computes
Water Demand	*Calculators ➔ Plumbing ➔ Water Demand*	Calculates daily domestic cold water & hot water storage demand (Liters/Day) per capita.
Tank Sizing	*Calculators ➔ Plumbing ➔ Tank Sizing*	Determines overhead and underground domestic water tank dimensions (Length x Width x Depth).
Sump Pump Sizing	*Calculators ➔ Plumbing ➔ Sump Pump*	Calculates drainage sump pit volume and pump discharge head/flow rate (GPM & Head Feet).
Sewage Pump Duty	*Calculators ➔ Plumbing ➔ Sewage Pump*	Sizes sewage lift station pumps and solids-handling impellers.
Loading Unit (BS 6700)	*Calculators ➔ Plumbing ➔ Loading Unit*	Converts fixture loading units (LU) to peak design flow rate (L/s) using Hunter's Curve.
Fixture Unit (IPC)	*Calculators ➔ Plumbing ➔ Fixture Unit*	Computes Water Supply Fixture Units (WSFU) and Drainage Fixture Units (DFU) for pipe sizing.
Gas Tank Sizing	*Calculators ➔ Plumbing ➔ Gas Tank*	Sizes LPG/Natural Gas storage tanks and vaporizers for commercial kitchens.
Hazen-Williams Pipe Loss	*Calculators ➔ Plumbing ➔ Hazen-Williams*	Computes friction head loss (PSI / 100 ft) for water supply and sprinkler piping.
Sprinkler Density	*Calculators ➔ Plumbing ➔ Sprinkler Density*	Computes NFPA 13 hazard classification flow density (GPM/sq ft) and remote area coverage.
Sprinkler K-Factor	*Calculators ➔ Plumbing ➔ K-Factor*	Calculates sprinkler discharge flow $Q = K \cdot \sqrt{P}$ for K-5.6, K-8.0, and K-11.2 heads.
Fire Water Demand	*Calculators ➔ Plumbing ➔ Fire Water Demand*	Computes total fire protection water storage volume for hydrants and sprinklers.
Fire Tank Duration	*Calculators ➔ Plumbing ➔ Fire Tank Duration*	Calculates required fire water tank retention time (30, 60, 90, or 120 minutes).
Fire Pump Duty	*Calculators ➔ Plumbing ➔ Fire Pump*	Determines main electric, diesel standby, and jockey pump flow rate (GPM) and total dynamic head (PSI).
FM200 Fire Suppression	*Calculators ➔ Plumbing ➔ FM200*	Calculates clean agent FM200 / Novec 1230 gas cylinder quantity for server room fire protection.

C. Electrical Tab

Calculator Name	Where to Click	What it Calculates & Computes

Cable Voltage Drop	*Calculators → Electrical → Cable Voltage Drop*	Calculates percentage voltage drop (% VD) and selects copper/aluminum cable cross-section (sq mm).
Busbar Voltage Drop	*Calculators → Electrical → Busbar Voltage Drop*	Computes sandwich busduct voltage drop and current carrying capacity.
Conductor Sizing	*Calculators → Electrical → Conductor Sizing*	Sizes phase, neutral, and earth conductors based on ambient temperature and grouping derating factors.
Circuit Breaker Sizing	*Calculators → Electrical → Circuit Breaker*	Selects MCB, MCCB, and ACB trip ratings (Amps) and short circuit withstand ratings (kA).
Transformer Sizing	*Calculators → Electrical → Transformer Sizing*	Calculates required kVA transformer rating with diversity and 20% future expansion margin.
Generator Sizing	*Calculators → Electrical → Generator Sizing*	Sizes standby diesel generator kVA / kW rating considering motor step starting kVA.
UPS Sizing	*Calculators → Electrical → UPS Sizing*	Calculates UPS rating (kVA) and battery bank backup runtime duration (Minutes).
Central Battery Sizing	*Calculators → Electrical → Central Battery*	Sizes central emergency lighting battery inverter capacity (3-hour backup per emergency code).
Capacitor Bank Sizing	*Calculators → Electrical → Capacitor Bank*	Computes reactive power (kVAr) required to improve power factor from 0.80 to 0.98.
LPD Calculation	*Calculators → Electrical → LPD Calc*	Computes Lighting Power Density (Watts / sq m) for ECBC & ASHRAE 90.1 compliance.
Lighting Lux Calculation	*Calculators → Electrical → Lux Calc*	Calculates number of LED luminaires required using the Lumen / Cavity Ratio method.
Short Circuit Calculation	*Calculators → Electrical → Short Circuit*	Computes 3-phase symmetrical fault current (kA) at main switchboards (MDB / SMDB).

👥 4. Role-Based Workflow & Task Matrix

🏢 Organization Admin (ORG_ADMIN)

* **Primary Responsibilities**: 1. **Manage License Seats**: View total available user seats (max_users) vs used seats (used_users). 2. **Invite & Manage Staff**: Add design engineers to your organization and grant appropriate roles. 3. **Configure Backup Paths**: Click *Organization* ➔ *Backup Settings* to set local (C:\Backups) and network server backup paths. 4. **Create Design Teams**: Go to *Teams* ➔ *Create Team* to group engineers into discipline teams (HVAC Team, Electrical Team). 5. **Review Audit Logs**: Click *Dashboard* ➔ *Activity Logs* to monitor organization activity history.

👷 Senior Design Engineer (Sr. Design Engineer)

* **Primary Responsibilities**: 1. **Project Creation & Management**: Create new engineering projects, enter client names, and set project codes. 2. **Assign Engineers & Teams**: Open *Project Modal* ➔ *Assign Members* to grant Junior Engineers access to specific projects. 3. **Version Approval**: Review saved calculation snapshots (V1.0 , V2.0) and restore baseline versions if needed. 4. **Export Master Schedules**: Generate combined MEP load schedules and export full project Excel reports.

🔧 Junior Design Engineer (Jr. Design Engineer)

* **Primary Responsibilities**: 1. **Perform Day-to-Day Calculations**: Input CFM, GPM, Lux, and Cable parameters into calculator sheets. 2. **Save Calculation Snapshots**: Click **"Save Version"** in any calculator to store named snapshots (e.g. *V1.0 - Initial HVAC Sizing*). 3. **Edit Assigned Projects**: Access and update calculation workspaces assigned by Senior Engineers.

👁️ Viewer (Viewer)

* **Primary Responsibilities**: 1. **Read-Only Inspection**: Inspect project calculations, space parameters, and single-line diagrams. 2. **Export PDF Reports**: Click **"Export PDF"** to print calculation sheets for client reviews or municipality submission.

📜 5. How to Save & Restore Calculation Versions (Version Control)

Version Control allows engineers to save named snapshots of their calculation sheets (e.g. *V1.0 - Concept Design*, *V2.0 - Final Client Submission*) and restore any past version with 1 click.

```mermaid
flowchart LR
    Step1["1. Open Calculation Sheet"] Step2["2. Click 'Save Version Snapshot'"]
    Step3["3. Enter Tag (V1.0) & Notes"] Step4["4. Open Version History"] Step5["5. Click 'Restore Version'"]

    Step1 --> Step2 --> Step3 --> Step4 --> Step5
```

💾 A. How to Save a Calculation Version Snapshot

1. Open any calculator (e.g., *HVAC Duct Sizer*, *Plumbing Water Demand*, or *Electrical Cable Voltage Drop*). 2. Enter your calculation inputs (CFM, Pipe GPM, Voltage, Cable Length, Lux). 3. Click the purple **"Save Version Snapshot"** button at the top-right of the calculation workspace card. 4. In the Version Popup: - **Version Tag**: Enter a version identifier (e.g. `V1.0`, `V1.1`, or `V2.0-Final`). - **Revision Description**: Add notes (e.g. *Updated duct friction factor per ASHRAE 2024*). 5. Click **"Save Version"**. 6. The snapshot is saved to local storage (`data/project_versions.json`) AND synchronized to Supabase Cloud DB (`project_versions`).

🕒 B. How to View Version History & Timeline

1. Inside any calculator workspace, click the **"Version History"** tab (or click **"View History"**). 2. A chronological timeline of all saved calculation snapshots is displayed, showing: - **Version Tag** (e.g. `V1.0`, `V1.1`) - **Author Name** (e.g. *John Doe - Sr. Design Engineer*) - **Timestamp** (e.g. *24 Aug 2026, 11:40 AM*) - **Revision Notes** (e.g. *Initial Duct Sizing for Tower A*)

🔄 C. How to Restore a Historical Version Snapshot

1. Open the **Version History Timeline**. 2. Locate the historical version you want to restore (e.g. `V1.0`). 3. Click the green **"Restore Version"** button next to that version item. 4. Click **"Confirm Restore"** on the confirmation popup. 5. **Instant Hydration**: Your current workspace inputs and calculation outputs immediately revert to the exact parameters of that historical version! 6. A notification banner displays: "*Restored to Version V1.0*".

📑 6. Generating & Exporting Engineering Reports (PDF & Excel)

1. **How to Export PDF Reports**: - Open any calculator (e.g. *Duct Sizer* or *Cable Voltage Drop*). - Complete your calculation inputs. - Click the blue **"Export PDF"** button at the top-right of the calculation card. - A branded calculation report with company header, input summary, and output results is saved to your downloads.

2. **How to Export Excel Schedules**: - Navigate to *Projects* ➔ *Select Project* ➔ *Export Load Schedule*. - Select desired tabs (HVAC, Plumbing, Electrical). - Click **"Export to Excel (.xlsx)"**. An editable multi-sheet workbook will be downloaded.